Analyzing Traffic Incidents in Izmir

Efe Korhan Uz
Industrial Engineering
Izmir University of Economics
Izmir, Turkey
efe.uz@std.ieu.edu.tr

Hüseyin Atacan Akgün
Software Engineering
Izmir University of Economics
Izmir, Turkey
huseyin.akgun@std.ieu.edu.tr

Hüseyin Yontar
Software Engineering
Izmir University of Economics
Izmir, Turkey
huseyin.yontar@std.ieu.edu.tr

***Abstract*—The abstract goes here.**
Our code for this project is available at [1].

I. Introduction

Traffic incidents are a prevalent problem in Izmir, imposing substantial costs on both citizens and the municipality, including life-threatening risks, economic losses, and operational burdens. Providing concise, data-driven evidence can accelerate decision-making and lead to precautionary actions that lower incident rates. While national studies exist, recent work centered on Izmir's incident patterns is scarce. This study analyzes a dataset of traffic incidents from the Izmir metropolitan area.

In this study, the incident records from the Izmir Municipality Open Data Portal are analyzed, including timestamps, location data and incident information. Our goal is to determine traffic incident patterns, summarizing locations and times at which risk concentrates across the metropolitan area.

Fig. 1. Map of the Izmir metropolitan area.

II. Background and Related Work

Numerous studies have investigated the spatial and temporal relationships of traffic incident occurrence using various analytical and data-driven methods.

[1]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir

In a study conducted by Cakıcı on Izmir [1], it is shown that the highest number of accidents in the years between 2020-2023 occurred in the autumn and summer seasons. In addition, least number of traffic accidents occurred during 2020 spring season due to the COVID-19 pandemic.

Another study conducted by Haybat and Karakas [2] reveals that incidents are focused on the high mobility regions of Izmir, especially in the south of Karsıyaka district, all over district of Konak, northeast of Karabaglar district, and northwest of Buca. Furthermore, the study reveals that while the number of incidents increased between 2010 and 2014, the total coverage area declined, suggesting a higher spatial density of incidents in later years.

According to a study performed by Kumar and Toshniwal [3] on road accidents in Gujarat, the time periods between 20:00 to 22:00 and 22:00 to 00:00 were identified as the most accident-prone invervals when the 24-hour day was divided into two-hour periods.

Cardelino (1998) [4] analyzes hourly traffic-counter data and reports the familiar weekday rush-hour double peak, contrasted with a relatively flatter weekend profile. Although framed around emissions, the study highlights the value of distinguishing traffic pattern types in time-of-day analysis.

Retallack and Ostendorf [5] report that crash frequency increases non-linearly as congestion rises. Their models, estimated on over five million hourly observations from 120 intersections, show that higher congestion is associated with disproportionately larger gains in expected crash counts, especially at the most congested levels. In short, the results indicate a positive, non-linear correlation between traffic congestion and crash rates.

III. Data Set Description

The data set used in this study was provided by the Open Data Portal of Izmir Metropolitan Municipality and authored by Izmir Transportation Center Directorate (IZUM) [6]. The data set includes detailed records of vehicle breakdowns and traffic accidents that occurred within the metropolitan area of Izmir. The metropolitan area of Izmir comprises the central districts of Çiğli, Karşıyaka, Bayraklı, Bornova, Konak, Buca, Karabağlar, Gaziemir, Balçova and Narlıdere, as illustrated in Figure 1.

At the time of our access to the data set, October 11, 2025, the observations cover the period between December 1, 2021,

TABLE I
DESCRIPTION OF EACH FEATURE IN THE RAW DATA SET.

Attribute	Type	Mean	Median	Mode	Number of Distinct Values	Number of Missing Values	Minimum	Maximum
Date	Interval	-	2023-07-18	2022-04-29	1397	0	2021-12-01	2025-09-28
Street	Nominal	-	-	Yeşildere Caddesi	183	0	-	-
Direction	Nominal	-	-	Merkez	157	16	-	-
Location	Nominal	-	-	Hilal Köprü Üstü	2032	14	-	-
Incident Type	Nominal	-	-	Maddi Hasarlı	30	0	-	-
Incident Time	Interval	-	15:30	17:29	1344	0	00:00	23:59
Intervention Time	Interval	-	15:52	18:10	1295	5193	00:00	23:59

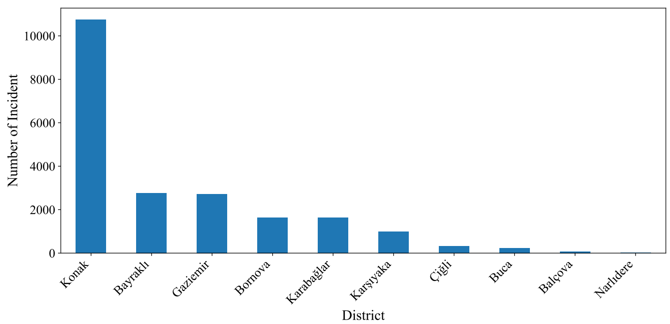

Fig. 2. Number of incidents per district between 2021-12-01 and 2025-09-28.

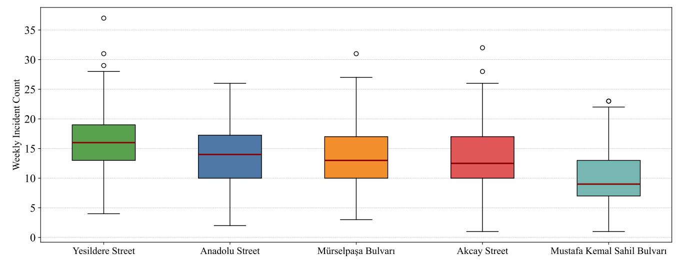

Fig. 3. Weekly incidents box-plot for top 5 most incident-dense street

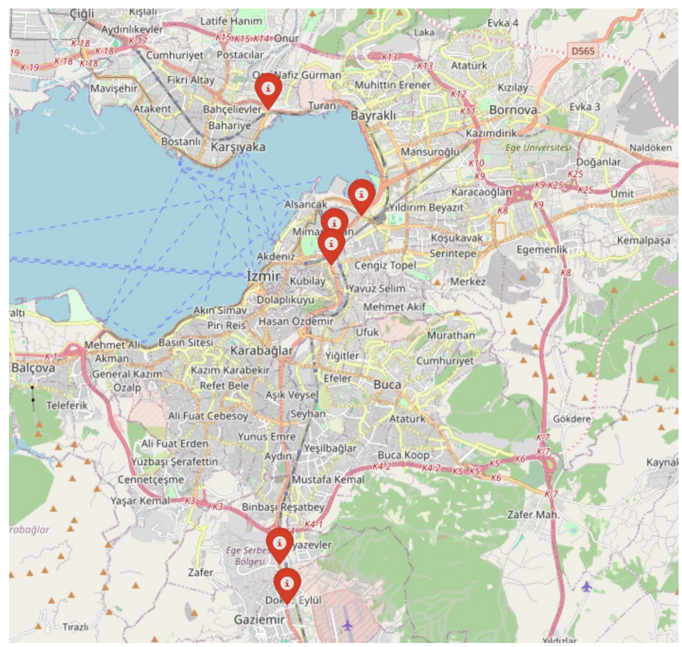

Fig. 4. Demonstrating Hotspot Incident Points of Izmir

and September 28, 2025. The raw data set consists of 21161 samples and 7 attributes. Each sample contains the street name, the direction of travel with 16 missing values, the location description with 14 missing values, the type of incident as nominal data type and the date of the incident, the accident time, and the intervention time with 5193 missing values as interval data type.

The descriptive information regarding the data set including mode and median values can be found in tabular form in Table I.

IV. PREPROCESSING

As part of the preprocessing stage, the dataset was examined to understand spatiotemporal incident patterns across Izmir's metropolitan districts and its most incident-dense streets.

Figure 2 demonstrates the total number of incidents by metropolitan districts. The bar chart shows that the highest number of incidents occurred in Konak, followed by Bayraklı and Gaziemir. Bornova, Karabağlar, and Karşıyaka also contribute considerably to the incident counts. In contrast, Çiğli, Buca, and Balçova contribute only a small share. The bar chart reveals that the total number of incidents in Konak is higher than the combined total of all other districts, indicating that Konak is the most incident-dense district in the Izmir metropolitan area.

Figure 3 illustrates the box plot of the weekly incident counts for the five most incident-dense streets of Izmir. From the box plot, Yesildere Street has the highest median weekly incident count, indicating a higher typical number of incidents than the other streets. Additionally, both Akçay Street and Mustafa Kemal Sahil Boulevard display lower minimum weekly incident counts compared to the other streets. Another notable pattern is that outliers are observed only on the upper end of the distributions, with no lower-end outliers present, indicating a right-skewed pattern in weekly incident counts.

Figure 4 displays the most incident-dense locations in Izmir: Mürselpaşa Boulevard on Hilal Bridge overpass with 954 incidents, Yeşildere Avenue on Tepecik Bridge overpass with 949 incidents, Mürselpaşa Boulevard on DGM Bridge overpass with 886 incidents, Akçay Avenue inside the Free Zone underpass with 521 incidents, Akçay Avenue inside the Gaziemir underpass with 497 incidents, and Anadolu Avenue after Naldöken Bridge with 484 incidents. This figure is crucial for pinpointing hotspots, where focused interventions produce the greatest impact per unit of effort and cost.

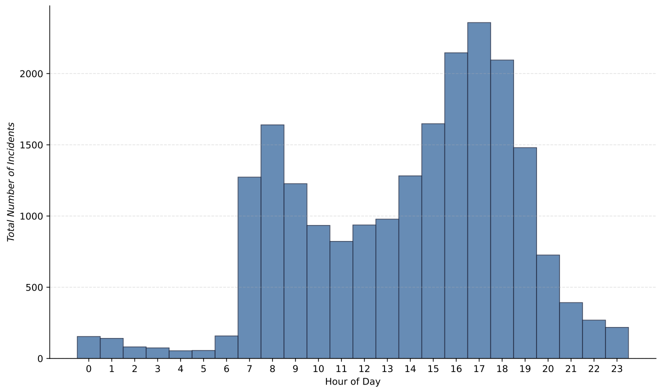

Fig. 5. Hourly incidents histogram

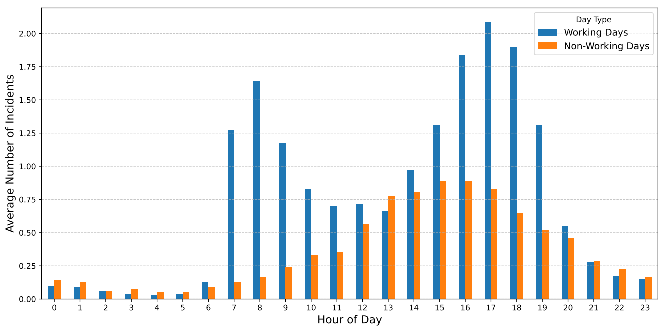

Fig. 6. Hourly incidents bar chart comparing working days and non-working days.

Figure 5 illustrates the distribution of incident counts across the hours of the day. The number of incidents exhibits an upward trend during the early morning hours, reaching its first peak at around 08:00. After this point, incident occurrences decline until midday. Following noon, incident counts rise again, with the highest rates observed around 17:00, which aligns with the end of typical working hours. Incident counts then decline sharply throughout the evening and continue to decrease into the late-night period.

Figure 6 compares the average hourly incident counts on working and non-working days. On working days, incidents begin to increase after 07:00, reach a morning high at 08:00, and then decline toward 11:00. Thereafter, average incident count gradually rise again and reaches its peak at 17:00. Following this point, incidents decline sharply until around 20:00 and continue decreasing into the late hours.

On non-working days, including weekends and national holidays, total incident numbers rise steadily through the morning, peak around 16:00, and then gradually decline into the late-night hours.

By examining hourly incident patterns, weekdays exhibit a double peak that aligns with established weekday congestion signatures, while weekends show a flatter profile with a pronounced mid-afternoon high [4]. Retallack and Ostendorf [5] further show that expected crash counts rise non-linearly

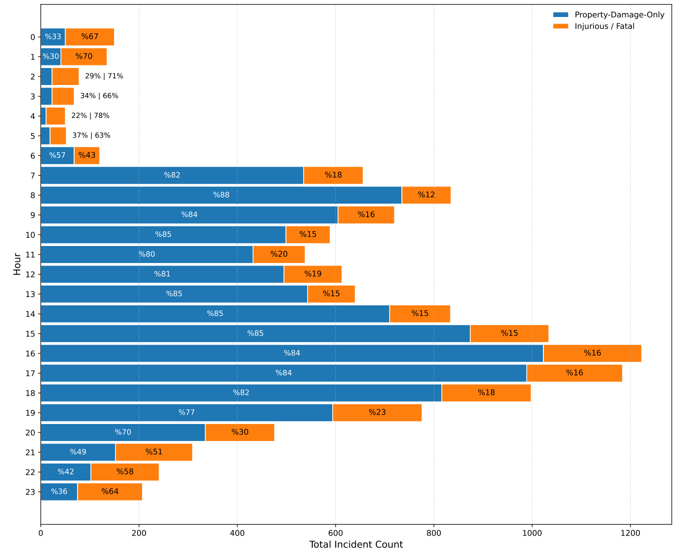

Fig. 7. Distribution of total accidents per hour including their accident type percentages

as congestion intensifies. According to their analysis, heavy traffic yields larger marginal increases in expected crashes, indicating a positive, non-linear congestion and crash rates.

Figure 7 shows how the distribution of accident types varies by hour of day, using a stacked bar chart of hourly totals with within-hour percentage shares. Although the number of accidents per hour decreases after 22:00, the analysis indicates that the ratio of injury or fatal crashes to property-damage-only incidents increases until 06:00. This figure suggests that while daytime traffic density contributes to high amount of incidents, nighttime incidents are observed to have smaller amount of incidents with higher injury and fatal share.

Figure 8 presents two bar charts: the top panel shows the number of distinct streets by district, and the bottom panel shows the average number of incidents per street. The first inference concerns Konak, across both panels, Konak ranks first in both street count and average incidents per street. Because overall incident totals are influenced by both the number of streets and the per-street average, Konak's leading total in Figure 2 is consistent with these results.

Another inference concerns Bayraklı. Figure 2 shows Bayraklı has the second-highest total incidents. In Figure 8, Bayraklı ranks sixth in street count and second in average incidents per street. This combination indicates very high incident density in Bayraklı, compensating for its smaller number of streets relative to other districts.

Finally, Buca ranks second in the number of distinct streets but eighth in average incidents per street. A similar pattern appears in Balçova, Narlıdere, and Çiğli as well, indicating lower incident density despite many distinct streets. This makes these districts relatively less important when prioritizing preventive interventions.

Figure 9 illustrates the relationship between the monthly number of distinct streets and total incidents, along with their

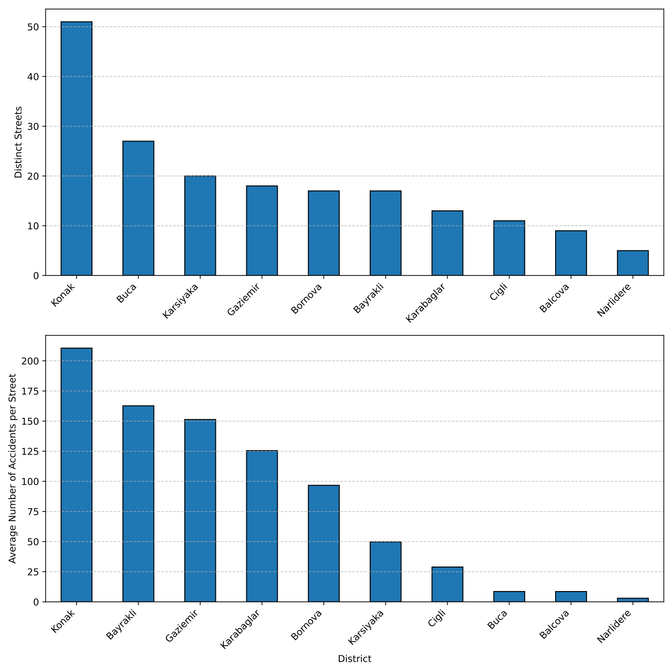

Fig. 8. Number of streets (top) and average incidents per street (bottom), by district

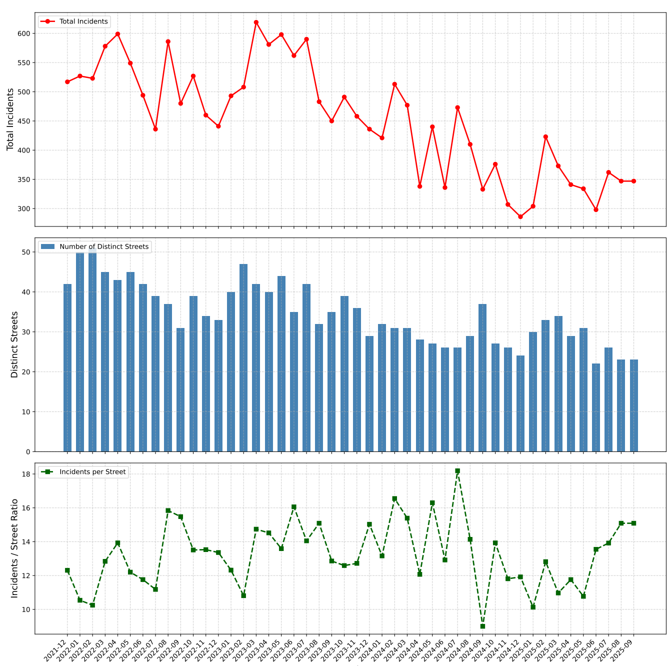

Fig. 9. Number of incidents as line graph (top), number of distinct streets as bar chart (middle), the ratio of total incidents/distinct street (bottom), by month

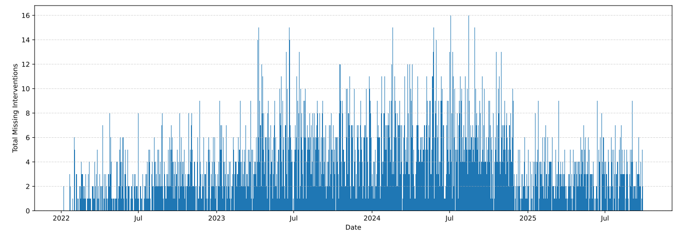

Fig. 10. Bar chart indicating the number of missing intervention time values in timeline

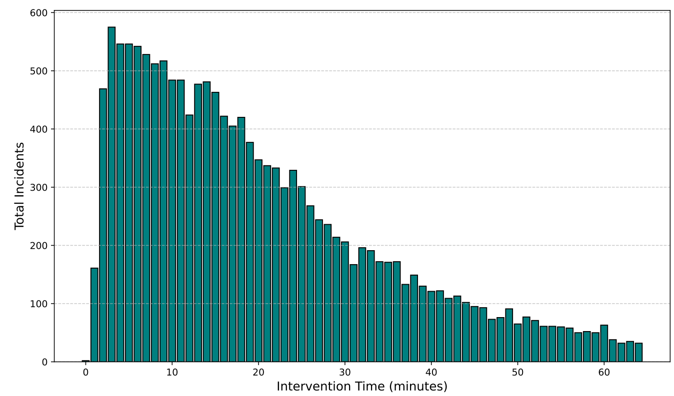

Fig. 11. Demonstrating total incidents in different intervention times

ratio, computed by dividing the total number of incidents by the number of distinct streets each month. The top panel shows a line graph of total monthly incidents, which gradually declines across the period. The middle panel displays a bar chart of the monthly count of distinct streets, which also gradually declines, following a trend similar to the line graph in the top panel. The bottom panel plots the ratio of total incidents to distinct streets. This series remains fairly steady, typically between 10 and 16, which aligns with the gradual decrease trend shown in the top and middle panels. In summation, the decrease in total incidents appears to align with a decrease in the number of distinct streets involved.

Figure 10 shows how many entries in the intervention time column are missing each day. There are in total 5193 missing entries in the raw data set. On inspection, the figure shows entries for each day, indicating that missing entries are uniformly distributed over the timeline.

Figure 11 shows the distribution of the duration between accident time and intervention time. The figure follows a right-skewed distribution, indicating that incidents with short intervention times are more common than those with long intervention times.

After examining the patterns in the missing intervention time entries, it is observed that there are no time periods where data was missing in bar blocks. In conclusion, it is decided to impute these missing intervention time values. To achieve this goal, first of all, the median duration between accident time and intervention time is calculated within detailed groups defined by accident type, location, time interval, street and

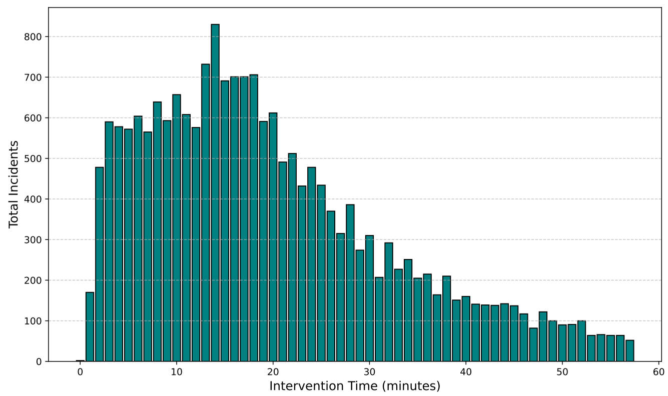

Fig. 12. Illustrating total incidents in different intervention times after imputing

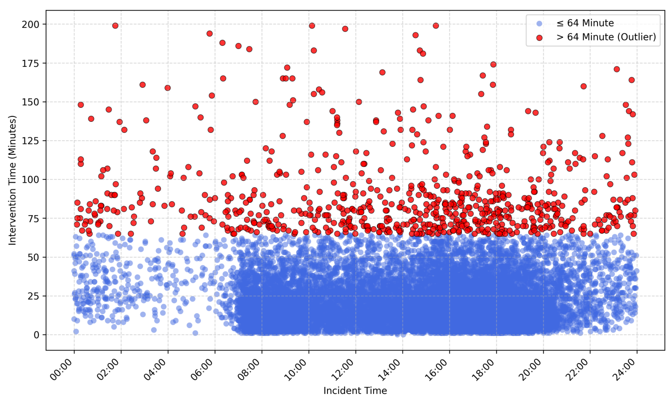

Fig. 13. Scatterplot Indicating Incident Time and Intervention Time Relations

working day status. Then, missing values are imputed. In the Figure 12, the distribution of intervention time with imputed values can be seen.

Figure 13 plots incident occurrence time against intervention time in minutes. Blue points show points with intervention times under 64 minutes and red points mark outliers with intervention times above 64 minutes. The scatterplot suggests that during high-traffic hours the outlier rate is lower, whereas during late-night and early-morning periods which are outside of regular working hours, display a higher outlier proportion, which could be explained by lack of available staff that could concern incidents.

Figure 14 provides heatmap, demonstrating rate of correlation between selected attributes. To construct the heatmap, we first selected the rows with the frequent incident types: vehicle breakdown, injury/fatal accident and property damage only accident, then partitioned the timeline into intervals with approximately equal incident counts. This procedure resulted in six bins with approximately 3400 incidents per bin: TIME_INTERVAL_A is between 20:00-7:54, TIME_INTERVAL_B is between 07:55-10:32, TIME_INTERVAL_C is between 10:33-14:23, TIME_INTERVAL_D is between 14:24-16:30, TIME_INTERVAL_E is between 16:31-18:01 and lastly

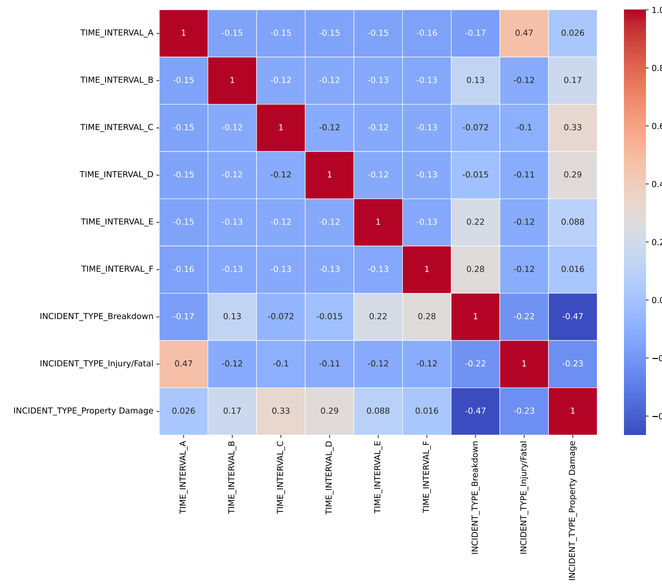

Fig. 14. Correlation heatmap of time intervals and

TIME_INTERVAL_F is between 18:02-19:59.

Then, indicator columns for frequent incident types were added: ACCIDENT_TYPE_Breakdown, ACCIDENT_TYPE_Injury/fatal and ACCIDENT_TYPE_PropertyDamage. Next, normalization process is applied to achieve reasonable results. Because these features are nominal, we applied one-hot encoding to obtain numeric representations, followed by normalization to conduct healthy comparisons across variables. Subsequently, encoded variables are multiplied with their respective total incident numbers.

The heatmap reveals several interesting patterns. ACCIDENT_TYPE_Injury/fatal shows a relatively high positive correlation number of 0.47 with TIME_INTERVAL_A. Since that interval comprises 20:00 to 07:54 hours, the injury incident rates are high as supported in Figure 7. Another observation is that for time intervals B, C and D the property damage incidents occurred more frequently, reaching correlation coefficient up to 0.33. Finally, ACCIDENT_TYPE_Breakdown is more common in intervals E and F with correlation coefficient reaching up to 0.28.

After discussing the figures, some additional findings about the attributes are discussed. In the GitHub repository of this project[2] and in Figure 14, there is a correlation matrix of some one-hot encoded categorical features. The purpose of this analysis is to identify potential relationships between different time intervals and incident types.

After searching for correlations between the attributes, in order to predict the type of the incidents, Random Forest Classifier and Decision Tree Classifier with Sequential Feature Selector were used for attribute selection. It is not ended with a

[2]https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/heatmap/onehot_heatmap.pdf

significant result when the selection done with Random Forest Classifier. For the Decision Tree Classifier, the best result was founded in the columns: Street, Direction, District, Season, isHoliday and Time Interval. The optimal subset was reached after 27 features with a best cross-validation accuracy of 0.5181. The result can be founded in the GitHub repository[3].

In the project's GitHub repository[4], we conducted Principal Component Analysis (PCA). First, we applied data-cleaning procedures and removed records with missing values. Next, we performed one-hot encoding to convert categorical variables to numeric form. To examine alternative correlation structures, we ran PCA on both the original Excel dataset and an edited version that included derived attributes such as intervention time, day type, and season. The resulting components did not yield informative structure for interpretation.

V. Classification / Regression

Describe the classification / regression algorithms that you have applied to your data set. Present and discuss your results

VI. Association Mining

Describe the association mining algorithms that you have applied to your data set Present and discuss your results

VII. Clustering

Describe the clustering algorithms that you have applied to your data set Present and discuss your results

VIII. Ensemble Learning

Describe the ensemble learning algorithms that you have applied to your data set Present and discuss your results

IX. Conclusion

Conclude your report by summarising your findings

Acknowledgment

The authors would like to thank...

References

[1] Z. Cakici, "Seasonal analysis of the distribution of urban traffic accidents: The case of İzmir province," in *5th International Conference on Innovative Academic Studies*. Konya, Türkiye: International Conference on Innovative Academic Studies, October 2024.

[2] H. Haybat and E. Karakaş, "An analysis of traffic accidents with spatial statistical methods in İzmir province," *Social Science Development Journal*, vol. 3, no. 13, pp. 599–617, October 2018, open Access, Refereed E-Journal. [Online]. Available: http://www.ssdjournal.org

[3] S. Kumar and D. Toshniwal, "Analysis of hourly road accident counts using hierarchical clustering and cophenetic correlation coefficient (cpcc)," *Journal of Big Data*, vol. 3, no. 1, jul 2016. [Online]. Available: https://journalofbigdata.springeropen.com/articles/10.1186/s40537-016-0046-3

[4] C. Cardelino, "Daily variability of motor vehicle emissions derived from traffic counter data," *Journal of The Air & Waste Management Association - J AIR WASTE MANAGE ASSOC*, vol. 48, pp. 637–645, 07 1998.

[5] A. E. Retallack and B. Ostendorf, "Relationship between traffic volume and accident frequency at intersections," *International Journal of Environmental Research and Public Health*, vol. 17, no. 4, 2020. [Online]. Available: https://www.mdpi.com/1660-4601/17/4/1393

[6] İzmir Büyükşehir Belediyesi, İzmir Ulaşım Merkezi Şube Müdürlüğü, "İzmir İli arızalı, kazalı araç verileri," Açık Veri Portalı, 2025, accessed on, October 11, 2025. [Online]. Available: https://acikveri.bizizmir.com/tr/dataset/izmir-ili-arizali-kazali-arac-verileri

[3] https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/blob/main/attribute_selection/sfs_decision_tree_accuracy_curve_without_cadde.pdf

[4] https://github.com/HuseyinYontar/Analyzing_Traffic_Incidents_in_Izmir/tree/main/pca